out of the 6 candidates. Including the redundant detection (1-1) results in high confidence scores for that match, whereas its exclusion leads to the 'None' class having the highest score. **d**, Inter-frame MLP example for two target cells: target 2 (not dividing) and target 3 (undergoing division). Each is associated with 7 nearest candidates in the next frame. **e**, Inter-frame MLP outputs using three different candidate selection sets (e.g., 1-5, 2-6, and 3-7) for target cells 2 and 3. Inclusion of the correct corresponding cell (e.g., 2-1 for target 2) or daughter cells (3-1 and 3-2 for target 3) in the candidate set results in high similarity scores from the MLP. Exclusion of true matches causes the 'None' score to dominate. The division prediction is mainly driven by the input division probability of the target cell, and is only marginally influenced by candidate selection. The visualization in (**a**) is based on the Cell Tracking Challenge (CTC) Fluo-N3DH-CE dataset, while the examples in (**b–e**) are from sequence #2 of the mskcc-confocal dataset. All results were generated using a model trained on sequence #3 of the mskcc-confocal dataset.